

Assignment 3

14X062 Computational Imaging

Task 1. Depth estimation

Depth estimation is the task of measuring the distance of each pixel relative to the camera. Depth is extracted from either monocular (single) or stereo (multiple views of a scene) images. Traditional methods use multi-view geometry to find the relationship between the images.

Based on \rf-depth-master\ssiDepthDemo.m investigate the particularities of depth estimation method for the images from Dataset1.

Ref: *rf-depth-master.zip*

Task 2. Image refocusing

Digital refocusing, a technique that generates photographs focused to different depths (distances from a camera) after a single camera shot, is attracting the attention of the computer graphics community and others in view of its interesting and useful effects.

Investigate properties of different image refocusing techniques:

1. \LightFieldRefocus\SelectiveRefocus\demoShiftSumRefocus.m
2. \LightFieldRefocus\SelectiveRefocus\demoSelectiveRefocus.m

Ref: *LightFieldRefocus.zip*

Task 3. Image refocusing and depth estimation

Investigate properties of image refocusing and depth estimation using images from OpEx and INRIA dataset.

Ref: <https://github.com/hahnec/plenoptacam>

Task 4 (optional, extra 1.0 point). Provide and investigate network-based image refocusing and depth estimation

Investigate the properties (image refocusing or depth estimation or both) of network-based system.

To complete all results (report together with your code) into archive and submit it to Computational imaging course moodle.